

Tendon Repair Rehab — Resident Quick Reference

For use at follow-up visits. Identify (1) what was repaired, (2) zone, (3) weeks post-op → cross-reference the protocol. Default protocols below follow BWH 2020 / 2007. Surgeon judgment overrides.

Which protocol?

Repair	Default protocol
FDS and/or FDP, zones 1–5, 4-strand core + epitendinous suture, surgeon clears EAM	Flexor EAM
FDS and/or FDP, zones 1–5, weaker repair, unreliable patient, nerve repair, or surgeon defers EAM	Modified Duran
FPL, any zone	FPL
Extensor, fingers (EDC/EIP/EDQ) — zones I–VIII	Primary Extensor (mallet / boutonniere / zone IV–VIII / SAM) → §4a–4d
Extensor, thumb (EPL) — zones T1–T5	Thumb Extensor → §4e
Extensor, wrist (ECRL/ECRB/ECU) — zones VII–VIII	Primary Extensor zones IV–VIII → §4c

Always confirm: zone, suture configuration (strands), concomitant pulley repair, nerve repair, vessel repair — these change the plan.

1. Flexor (FDS + FDP) — Early Active Motion · Zones 1–5

Therapy frequency: 1–2x/week × 8–12 weeks. Orthosis day 3.

Phase	Orthotic	Exercises	Precautions
I — Immediate (day 3 to wk 2)	Zones 1–3: dorsal forearm-based; wrist 20° ext, MCP 30–40° flex, IP 0°. Zones 4–5: dorsal forearm-based; wrist 0°, MCP 60–75° flex, IP 0°. Add pulley ring if pulley repaired.	Passive DIP & PIP flex, active ext to orthosis. Passive MCP flex 60–80° with active IP ext. Passive wrist flex, active wrist ext to orthosis. Day 4–5: after passive, active motion to 25% fist + active ext to orthosis.	No passive wrist ext beyond 0° (zones 4–5) if median/ulnar n. repaired ×6 wk. Avoid place-and-holds (pulley buckling). Tendon weakest day 5–21.
II — Protective (wk 2–4)	Wk 4: transition orthosis to hand-based.	Active to 50% fist (wk 2), 75% fist (wk 3–4). Wk 3: light fine motor out of orthosis in clinic. Wk 4: tendon gliding + isolated FDS glide to repaired finger, wrist 0°–20°; light fine motor at home.	—
III — Intermediate (wk 4–6)	Gradually wean during day. Discharge orthosis by 6 wk. If IPs stiff at 6 wk: convert dorsal block → volar hand-based nighttime (zone 2) or volar forearm-based (zone 3).	Tendon gliding wrist 20–30° ext. Wk 5: DIP & PIP blocking. Wk 6: progress functional activities.	No composite wrist >30° + combined finger ext until 6 wk. Avoid PIP/DIP blocking of small finger (rupture risk). Weightlifting restricted.
IV — Minimal protection (wk 6–12)	D/C night orthosis when digital active ext = 0°.	Wk 8: light graded strengthening; resisted isolated DIP & PIP flex. Progress to work/sport per MD.	No resisted grip/pinch until 8 wk.

Visual cue: Wk 1–2 = touch IF (25% fist) · Wk 2–3 = touch LF (50%) · Wk 3–4 = touch RF (75%).

2. Flexor (FDS + FDP) — Modified Duran · Zones 1–5

Use when EAM is deferred. Same orthotic, exercises are passive only in Phase I.

Phase	Orthotic	Exercises	Precautions
I — Immediate (day 3 to wk 2)	Same as EAM (zones 1–3 vs 4–5). Pulley ring if pulley repaired.	Passive DIP & PIP flex, active ext to orthosis. Passive MCP block 60–80° with active IP ext. Passive composite fist. Passive wrist flex, active wrist ext. In clinic (orthosis off): passive wrist ext with fingers passively flexed; passive wrist flex with passive hook fist (prevent intrinsic tightness).	No passive wrist ext beyond 0° (zones 4–5) if nerve repaired ×6 wk. Avoid place-and-holds. Reduced blood flow in zone 1.
II — Protective (wk 2–4)	Wk 4: hand-based.	Wk 3: light fine motor out of orthosis in clinic. Wk 4: tendon gliding + isolated FDS glide, wrist 0°; light fine motor at home.	—
III — Intermediate (wk 4–6)	Wean during day. D/C by 6 wk. Convert to nighttime orthosis if IP stiffness.	Tendon gliding wrist 20° ext. Wk 5: DIP & PIP blocking. Wk 6: progress function.	No composite wrist >20° + combined ext until 6 wk. Avoid small-finger blocking.
IV — Minimal protection (wk 6–12)	D/C night orthosis when digital ext = 0°.	Wk 8: light graded strengthening, resisted isolated DIP/PIP flex.	No resisted grip/pinch until 8 wk.

3. Flexor Pollicis Longus — All Zones

Week	Orthotic	Exercises	Precautions
0–3	Dorsal blocking: wrist neutral, CMC flexed/abducted under 2nd MC, MP full ext. Zone I only: add separate dorsal gutter blocking IP at 30° flex.	Passive composite thumb flex / active ext to orthotic. Passive IP flex / active ext to orthotic. Gravity-assisted wrist flex / active ext. Tendon gliding digits 2–5. EAM (if cleared + 4-strand + epitendinous): place/hold thumb flex with wrist ext, start 48–72 hr post-op.	No active thumb flex unless cleared EAM. No passive wrist ext. No passive thumb ext. No functional use.
3	Same.	Add place/hold thumb flex with wrist passively extended (gentle). Continue prior.	Avoid co-contraction.
4	Convert to hand-based.	Active non-resistive thumb flex with wrist ext.	Light prehensile in therapy.
5	D/C orthosis.	Gentle blocking for thumb IP flex.	—
6	Dynamic IP ext splinting PRN.	Putty scraping PRN. May start NMES/US PRN.	—
8	—	Gradually add resistive exercises.	—

4. Primary Extensor — Hand & UE (2020)

4a. Mallet (Zone I — DIP) & Zone II (middle phalanx)

Phase	Orthotic	Exercises	Precautions
I — Immediate (day 1 to wk 6–8)	DIP 10°–0° hyperextension. Tendinous mallet: 6–8 wk. Bony mallet: 6 wk. Op: 100% × 6 wk. Provide 2 splints (one for showering).	Active PIP flex of affected finger with adjacent fingers held in extension.	Daily skin check, DIP must stay extended. Consider taping DIP. If swan-neck → reduce passively, dorsal block PIP at 30°.
II — Protective (wk 6 bony / wk 8 tendinous)	Tendinous: orthosis 100% except exercise/hygiene. Bony: orthosis for strenuous activity + sleep × 2–4 wk.	Remove orthotic. Gentle AROM DIP flex/ext: start 10° flex, progress 10° per week. Replace orthotic. Wk 8: light activity out of orthosis if no lag.	If DIP extensor lag ≥10° → orthosis 100% × 2–4 wk.
III — Intermediate (wk 10)	D/C during day. Night orthosis × 2 more wk.	Fine motor, gradual flexion while maintaining DIP ext.	Most zone 1–2 injuries finish with –10–0° extensor lag.

4b. Boutonniere (Zone III — PIP)

Phase	Orthotic	Exercises	Precautions
I — Immediate (day 1 to wk 6)	Orthosis or circumferential cast with PIP at 0°. Op: 100% × 6 wk.	Active DIP flex of affected finger.	If DIP hyperextension at wk 2 → reduce passively.
II — Protective (wk 6)	Convert to orthosis with PIP at 0°. Wk 7: reduce gradually as 0° PIP ext maintained. Light activity if no lag.	Gentle active PIP ext to 30° flex, progress 10° per week. Replace orthotic.	If PIP extensor lag ≥10° → orthosis 100% × 2–4 wk.
III — Intermediate (wk 10)	D/C.	—	—

4c. Zones IV–VIII (proximal phalanx through forearm) — Immobilization Protocol

Week	Orthotic	Exercises	Precautions
1–6	Volar digit static, PIP at absolute 0°, or serial cast. Lateral bands repaired → include DIP at 0°. Lateral bands intact → DIP free.	ROM may start wk 3–6 depending on healing. AROM PIP flex to 30°, progress 10–20°/wk if no lag, 10 reps hourly. Lateral bands repaired → tendon gliding wk 3. Intact → wk 1.	No forceful flex. No gripping. Splint between sessions. Serial cast if PIP contracture, closed injury, or noncompliant.
6–8	Wean from splint during day. Night splint continues.	AAROM or dynamic flexion splint may start. Composite wrist + digit flex.	Light functional use out of splint (≤1–3 lb).
10–12	D/C splint.	PROM / PREs.	—

4d. Zones III–IV Alternative — Short Arc Motion (SAM) Protocol

Consider when broad tendon-bone interface in zone IV and adhesion risk is high.

Week	Orthotic	Exercises	Precautions
1	Digit volar immobilization: PIP & DIP at 0°, 24/7 except exercise. Two volar exercise splints: template 1 (PIP 30° flex, DIP 20° flex), template 2 (PIP 0°, DIP free).	Remove immobilization hourly, AROM PIP & DIP, 10–20 reps in templates 1 & 2. Wrist held 30° flex, MP at 0°. Lateral bands repaired → limit DIP flex to 30–35° in template 2; intact → fully flex/extend DIP.	PIP must be at 0° in immobilization splint to prevent extensor lag.
2	If no lag: progress template 1 → PIP 40–50°, DIP 30–40°.	If extensor lag develops → smaller flex increments, focus on extension.	If rupture suspected → MD.
3	If no lag: template 1 → PIP 50–60°, DIP 40–50°.	—	—
4	If no lag: template 1 → PIP 70–80°, DIP 50–60°.	—	If PIP stiff → intermittent splint into flex, continue static ext splint into wk 5–6.
5	Begin weaning.	Composite flex and gentle PREs.	Light functional activities out of splint.
6	D/C splint. Night PRN.	PROM & PREs, reverse putty scraping.	—

4e. Thumb Extensor (EPL) — Zones T1–T5 (BWH 2007)

Thumb zones differ from finger zones. EPL has its own zone map (T1–T5) and rehab protocols. Do not use finger zone I–VIII protocols for EPL repairs.

Zone T1 (IP joint) — Immobilization

Week	Orthotic	Exercises	Precautions
1–3	IP at 0° or slight hyperextension, continuously. Op: 5–6 wk. Non-op: 8 wk. Provide 2nd splint for showers.	None.	No IP flexion. No gripping or pinching in splint. McConnell tape to hold digit during splint changes.
5–6	Op: remove for exercise, otherwise continuously x2–4 more wk.	Op: AROM IP flex in 20° increments/wk, modifying if extensor lag develops, 10 reps hourly. Non-op: no ROM yet.	If extensor lag $\geq 10^\circ$ → continuous splint x2–4 wk, reassess.
8	Gradually wean during day. Continue at night.	Op: AAROM if needed, provided no extensor lag. Non-op: initiate AROM IP flex in 20° increments.	—
10–12	D/C splint.	Op: PROM, PREs (light grip/pinch). Non-op: AAROM → PROM → PREs as tolerated.	—

Zone T2 (proximal phalanx) — Immobilization

Week	Orthotic	Exercises	Precautions
1	Hand-based static splint (short opponens): MP and IP at 0°, thumb in radial abduction.	None.	No active motion.
3	Same.	Initiate AROM flexion at each joint; progress 25–30°/wk.	Tendon-to-bone adherence may limit excursion in this zone.
4–5	Same.	AAROM flexion, isolated and combined joint. Light prehension ADL out of splint.	—
6	Begin to wean. Dynamic flexion splinting PRN.	Moderate prehension ADL out of splint.	—
8	D/C splint.	PREs. Full function.	—

Zones T3–T4 (MP joint / metacarpal) — Controlled Passive Motion

Week	Orthotic	Exercises	Precautions
1	Forearm-based, static or dynamic: thumb MP at 0° (not HE), slight abduction, wrist 30° ext. If dynamic → also fabricate static forearm-based night splint (wrist 30° ext, MP 0°).	Initiate AROM flexion in 20° increments/wk. PROM extension (via dynamic traction, self-PROM, or to static splint limit).	No active extension. No gripping or pinching, even in splint. Splint and exercise choices based on strength of repair, scarring potential, and patient.
2–3	Same.	Increase AROM flexion arc as tolerated. Place-and-hold extension may start wk 3.	—
4	Same.	AROM in extension.	—
5–6	Initiate dynamic flexion splinting PRN.	Full AROM flexion, isolated and combined.	—
6–8	D/C splint.	PREs.	—

Zone T5 (retinaculum / wrist) — Dynamic Extension Splinting

Week	Orthotic	Exercises	Precautions
1	Dynamic extension splinting as in T3–T4.	Same as T3–T4 wk 1.	Dense adhesions may limit EPL excursion at retinacular level. Proper wrist and thumb positioning crucial.
3	Same.	May initiate AAROM flexion.	—
5–8	Progress as T3–T4 wk 5 onward.	—	—

Red flags at follow-up

- **Sudden loss of active flexion / extension** after gain → assume rupture. Refer to surgeon.
- **Extensor lag $\geq 10^\circ$** → return to full-time orthosis × 2–4 weeks.

- **PIP contracture developing** → serial casting or static progressive splint, escalate therapy.
- **No improvement in tendon glide after 6–8 weeks** → discuss tenolysis (typically deferred to ≥3–6 months post-op).
- **Persistent edema beyond 4 weeks** → edema control, rule out infection.

Key principles

- Repair is weakest **day 5–21**.
- Strong repair (4-strand core + epitendinous) is the prerequisite for EAM.
- Adhesions favor extrinsic healing; controlled motion favors intrinsic healing.
- Gap >3 mm → ↑ rupture and adhesion risk.
- Always cross-check with surgeon's op note (suture configuration, pulley/nerve/vessel work).

Sources (BWH Department of Rehabilitation Services)

- Flexor EAM (2020) — <https://www.brighamandwomens.org/assets/BWH/patients-and-families/rehabilitation-services/pdfs/flexor-tendon-repair-hand-therapy-protocol.pdf>
- Modified Duran (2020) — <https://www.brighamandwomens.org/assets/BWH/patients-and-families/rehabilitation-services/pdfs/modified-duran-hand-therapy-protocol.pdf>
- FPL (2007) — <https://www.brighamandwomens.org/assets/BWH/patients-and-families/rehabilitation-services/pdfs/hand-fpl-repair-pt-protocol-bwh.pdf>
- Primary Extensor (2020) — <https://www.brighamandwomens.org/assets/BWH/patients-and-families/pdfs/upper-extremity-extensor-tendon-repair-protocol-2020.pdf>
- Primary Extensor incl. Thumb Zones (2007) — <https://www.brighamandwomens.org/assets/BWH/patients-and-families/rehabilitation-services/pdfs/hand-extensor-tendon-repair-protocol-pt-all-bwh.pdf>

BWH protocol index: <https://www.brighamandwomens.org/patients-and-families/rehabilitation-services/physical-therapy-protocols>

Compiled by Dr. Janos Barrera (NYU Langone) as a teaching aid. Always defer to the operating surgeon's specific instructions.